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(1) Research Program/Project/Study Title: ACCEPTABILITY OF BANGUS-CLAM SISIG	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	ROXAS CITY						
		Province:	CAPIZ						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☒ Basic	☐ Pilot Testing								
☐ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity Food processing sector Restaurants Small and medium enterprise Academe		(8) Discipline/Field of Specialization Culinary							

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(9) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Milk Fish is one of the healthy and nutritious fish available. The fish is loaded with omega 3 fatty acids and protein which is quite essential for proper functioning of the body. while Clams are rich in Vitamin B12, which provides iron and protein. Clams may help to regulate high blood pressure. It also improves thyroid health. They contain high amounts of riboflavin and also choline. Clams are rich in many minerals and vitamins but are low in fat. The study utilized Bangus flakes and clam meat of varying proportion as one of the ingredients with Butter, Garlic, Chili, Onion, Liver spread, Soy Sauce, Black Pepper, Annato Oil, Oil and Salt. Guided by the foregoing objectives : This study was conducted to determine the acceptability of Bangus-Clam Sisig, specifically it aimed to develop a bangus-mud clam sisig product, to determine the sensory characteristic of bangus-clam among treatment in terms of color, odor, taste and texture, determine if there is a significant difference on the acceptability of bangus-clam sisig among treatments, determine the most preferred formulation (MPF) of bangus-clam sisig among treatments, determine the consumer acceptability of the most preferred formulation. The development was done in 5 treatments with 3 replications. The quality of the product was determined by using sensory evaluation. The bangus-clam sisig had similar color, odor, taste texture and acceptability regardless of the amount of bangus flakes and clam meat added to the formulated product. The addition of bangus and clam in the develop sisig even of varying percentage had no intense effect on its overall results in terms of color, odor, taste and texture, however, during the cooking process, especially when pan-frying, the skin browns and crisps up. This contributes a significant amount of brown color to the overall dish. The result shows that there was no significant difference in the overall acceptability of bangus-clam sisig. Regardless the amount of bangus and clam sisig. All treatments are acceptable. Out of five treatments it shows that the most preferred formulation consists of 50% bangus and 50% clam. It shows that most of the college student preferred the bangus-clam sisig.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Bangus flakes, Seafood, Ready to eat, processed foods

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Milkfish *Chanos chanos*, locally known as “Bangus” is the top fish commodity in terms of production and consumption in the Philippines. (BFAR, 2022) Bangus can be raised anywhere. The top bangus producing provinces are Bulacan, Pangasinan, Capiz, Iloilo, and Negros Occidental. The most recent report released by the Bureau of Agricultural Statistics (BAS) show that the combined production of these five provinces alone accounts for more than 50 percent of the country’s total production. (Tacio, H., 2010) It is also one of the largest volumes of fish merchandized every day in almost all public markets in the country. (BFAR, 2022) Milk Fish is one of the healthy and nutritious fish available. The fish is loaded with omega 3 fatty acids and protein which is quite essential for proper functioning of the body. (Shadrack, R.S., Gevera, S., Pickering, T., and Ferreira, M., 2021) Most of the milkfish sold in local markets are in fresh chilled form either in whole or deboned. Nowadays, more of milkfish harvest is processed into value-added forms such as smoked, dried, marinated, fermented, canned or bottled. (BFAR, 2022)

Clams are popular shellfish that have a high nutritional density and are high in vitamins and minerals. Whether steamed, fried, or grilled, clams are a delicious and nutritious addition to your diet. Clams are bivalve mollusks, a form of shellfish with a shell. This shellfish (typically) resides on the sandy ocean floor and is found in different cuisines throughout the world. (Elvira, M. and Jumawan, J., 2017)

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Clams are nutritious with loads of health benefits. Clams are rich in Vitamin B12, which provides iron and protein. Clams may help to regulate high blood pressure. It also improves thyroid health. They contain high amounts of riboflavin and also choline. Clams are rich in many minerals and vitamins but are low in fat. Clams are considered one of the most nutritious foods all over the world. Clams contain omega-3 fatty acids, which are beneficial to maintain good heart health and also reduce inflammation. (Arora, R., 2021)

With the abundance of the mentioned raw materials - bangus and mud clam, the researcher would like to explore the utilization of the commodity in the production of sisig. Sisig is a Filipino dish that is made from parts of pig head and chicken liver. Sisig is usually seasoned with calamansi, onions, and chili peppers. Sisig was first mentioned in a Kapampangan dictionary in the 17th century, which means “to snack on something sour” and “salad.” (Valenzuela, K. R., Rodriguez, J., Guzman, R. S. ., Calades, J. M. ., Dela Cruz, R. ., Abesia, J. ., Nawal Amir, & Tolato, A. L. ,2020) The product will add market value to the raw materials as well as generate income for the community where the technology would be transferred.

This study was conducted to determine the acceptability of Bangus-Mud Clam Sisig, specifically it aimed to develop a bangus-mud clam sisig product, to determine the sensory characteristic of bangus-clam among treatment in terms of color, odor, taste and texture, determine if there is a significant difference on the acceptability of bangus-clam sisig among treatments, determine the most preferred formulation (MPF) of bangus-clam sisig among treatments, determine the consumer acceptability of the most preferred formulation.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Materials and Methods

Materials.

The following are the materials used in the preparation of Bangus-Clam Sisig: Knife, Chopping Board, Preparation Bowl, Weighing Scale, Pot & pans, Frying Pan, Ladle.

Raw materials

The following are the raw materials used in the preparation of Bangus-Clam Sisig: Bangus, Mud Clam, Butter, Garlic, Chili, Onion, Liver spread, Soy Sauce, Black Pepper, Annato Oil, Oil and Salt.

Experimental Treatment

Ingredients	Treatment A	Treatment B	Treatment C	Treatment D	Treatment E
Bangus	200g	175g	150g	100g	0
Mud Clam	0	25g	50g	100g	200g
Oil	320ml	320ml	320ml	320ml	320ml

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Annatto Oil	30ml	30ml	30ml	30ml	30ml
Butter	20g	20g	20g	20g	20g
Liver spread	15g	15g	15g	15g	15g
Garlic	15g	15g	15g	15g	15g
Onion	15g	15g	15g	15g	15g
Mayonnaise	15g	15g	15g	15g	15g
Soy Sauce	15g	15g	15g	15g	15g
Chili	5g	5g	5g	5g	5g
Salt	5g	5g	5g	5g	5g
Black Pepper	3g	3g	3g	3g	3g

METHODS

Collection of Bangus and Clam

Fresh bangus and Clam were collected from the pond of CapSu Dayao, Roxas City and bring to the area of Fisheries and Food Research Development Center (FFRDC) for deboning and depuration of clams.

Preparation of Bangus

1. Washed the boneless bangus in running water to remove the excess blood and adhering dirt.
2. Seasoned the boneless bangus in salt and Black pepper
3. Preheated the veg. oil to a temperature between 175F to 190F
4. Fry the boneless bangus in temperature of 145 F (63 C)
5. Set aside the cooked boneless bangus until cool
6. Flakes the boneless bangus
7. Set aside

Preparation of Mud Clam

1. Brush the clam shell and washed thoroughly in running water to remove the adhering dirt.
2. Soak the clams in water with chili for 3hrs.
3. Washed thoroughly in running water, until totally clean
4. Steam the clam for 20mins until the clam shells are wide open
5. Let it cool
6. Remove clam meat from the shell
7. Remove the intestine of the clam
8. Chop the meat clam
9. Set aside

Cooking Procedure of Bangus-Clams Sisig

1. Heat the annatto oil over low fire
2. Add Butter
3. Sauté garlic until light brown
4. Caramelized onion

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5. Add clams, stir until tender
6. Add bangus flakes
7. Add remaining ingredients; liver spread, soy sauce, chili, salt, black pepper and mayonnaise.
8. Mixed well and taste
9. When done, remove from the heat.
10. Cool completely before storing in vacuum pack

Sensory Evaluation of Bangus- Clam Sisig

Sensory evaluation consists of judging the quality of food by a panel of 10 food experts. The evaluation deals with analyzing and interpreting the qualities of bangus- mud clam sisig as they are perceived by the sense of sight, taste, touch, Category scale was used to determine the sensory characteristics of bangus-clam sisig develop. Sensory attributes of the bangus-clam sisig namely, color, odor, taste and texture, consumer acceptability consist of 100, (50 students and 50 professional). The scoring scale was: 1 (dislike extremely), 2 (dislike very much), 3 (dislike moderately), 4 (dislike slightly), 5 (neither like or dislike), 6 (liked slightly), 7 (liked moderately), 8 (liked very much), 9 (liked extremely).

Experimental Design and Treatment

Experimental Design

The study used experimental method of research. An experimental method is a method or procedure involving the control or manipulation of conditions for the purpose of studying the relative effects of various treatments applied to the mixture of the different samples (Calderon, 2002).

Experimental Treatment

The study used five (5) treatments with three replications:

Treatment A (100% bangus, no clam), Treatment B (75% bangus, 25% clam), Treatment C (50% bangus, 50% clams), Treatment D (25% bangus, 75% clams) Treatment E (100% clam, 0 bangus) The same ingredients and proportion was used in the study except for the proportion of bangus and clam which vary per treatment.

Data Collection

The data was gathered using score containing the 5 category scale for sensory and 9-point hedonic scale for acceptability. The data was gathered after every replication.

Scoring of Variables

The criteria for evaluation as to sensory characteristics of the product was 5 category scale in terms of color, odor, taste and texture. The rating for general acceptability in 9-point hedonic scale was: 9 for like extremely, 8 for like very much, 7 for like moderately, 6 for like slightly, 5 for neither like nor dislike, 4 for dislike moderately, 3 for dislike slightly, 2 for dislike very much and 1 for dislike extremely (Calmorin 2006).

The score on each item was interpreted based on the result of the computation as to color, odor, taste, texture and general acceptability using the following scoring interval and verbal interpretation.

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For Color

Scoring	Verbal Interpretation/ Response Categories	Scoring Interval
5	Brown	4.2-5.0
4	Light brown	3.4-4.9
3	Slightly brown	2.6-3.39
2	Moderately brown	1.8-2.59
1	yellow	1.0-1.79

For Odor

Scoring	Verbal interpretation/ Response Categories	Scoring Interval
5	No fishy odor	4.2-5.0
4	With slight fishy odor	3.4-4.9
3	With slightly strong fishy odor	2.6-3.39
2	Moderately strong fishy odor	1.8-2.59
1	Extremely strong fishy odor	1.0-1.79

For Texture

Scoring	Verbal Interpretation/ Response Categories	Scoring Interval
5	Crispy	4.2-5.0
4	Slightly crispy	3.4-4.9
3	soft	2.6-3.39
2	Fairly chewy	1.8-2.59
1	Very chewy	1.0-1.79

For Taste

Scoring	Verbal interpretation/ Response Categories	Scoring Interval
5	No bangus-clam taste	4.2-5.0
4	With slight bangus-clam taste	3.4-4.9
3	With slightly strong bangus-clam taste	2.6-3.39
2	Moderately strong bangus-clam taste	1.8-2.59
1	Extremely strong bangus-clam taste	1.0-1.79

For Acceptability

Scoring	Score Interval	Verbal interpretation/Response Categories
9	8.12 – 9.0	Like Extremely
8	7.23 – 8.11	Like Very Much
7	6.34 – 7.22	Like Moderately

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6	5.45 – 6.33	Like Slightly
5	4.56 – 5.44	Neither Like nor Dislike
4	3.67 – 4.55	Dislike Slightly
3	2.78 – 3.66	Dislike Moderately
2	1.89 – 2.77	Dislike Very Much
1	1.00 – 1.88	Dislike Extremely

Statistical Tools and Analysis

The data was analyzed using weighted mean and Analysis of Variance (ANOVA).

The weighted mean was used to determine the acceptability of the sensory characteristics such as color, odor, taste, texture and general acceptability of the product.

The analysis of variance was used to determine the significant difference among treatment in terms of color, odor, taste, and texture.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Results and Discussion

Results and Discussion

Sensory attributes of Different Treatments in terms of color, odor texture and taste

In terms of color

Table 1 shows the result of the sensory attributes of different Treatments in terms of color.

As shown in the Table, Treatment A, had a mean of 3.57, interpreted as light brown; Treatment B had a mean of 3.37, interpreted as slightly brown, Treatment C had a mean of 3.57 interpreted as light brown, Treatment D had a mean of 3.53, interpreted as light brown and treatment E had a mean of 3.47 interpreted as light brown.

Bangus or milkfish, has a naturally grayish-silver skin. But during the cooking process, especially when pan-frying, the skin browns and crisps up. This contributes a significant amount of brown color to the overall dish.

Table 1. Sensory attributes of different treatments in terms of color.

Treatment	Replicate	Total	Mean	Verbal Interpretation
	1	2	3	

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A	3.7	3.3	3.7	10.7	3.57	Light brown
B	2.8	3.5	3.8	10.1	3.37	Slightly brown
C	3.3	3.9	3.5	10.7	3.57	Light brown
D	3.5	3.6	3.5	10.6	3.53	Light brown
E	3.6	3.5	3.3	10.4	3.47	Light brown

In terms of odor

Table 2 shows the result of the sensory attributes of different treatments in terms of odor.

As shown in the Table, Treatment A had a mean of 3.97, Treatment B had a mean of 3.37, Treatment C had a mean of 3.57, Treatment D had a mean of 3.40 and treatment E had a mean of 3.57 all interpreted as with slight fishy odor.

The bangus will contribute a subtle, fresh seafood aroma. It only shows that regardless of the amount of bangus and the amount clam added to the formulated product the odor is still similar.

Table 2. Sensory attributes of different treatments in terms of odor.

Treatment	Replicate			Total	Mean	Verbal Interpretation
	1	2	3			
A	4.1	3.7	4.1	11.9	3.97	With slight fishy odor
B	3.2	4.0	4	11.2	3.73	With slight fishy odor
C	3.7	3.2	3.8	10.7	3.57	With slight fishy odor
D	3.5	3.4	3.3	10.2	3.40	With slight fishy odor
E	3.7	3.7	3.3	10.7	3.57	With slight fishy odor

In terms of Taste

Table 3 shows the result of the sensory attributes of different treatments in terms of taste.

As shown in the Table, Treatment A had a mean of 3.77, Treatment B had a mean of 3.57, Treatment C had a mean of 3.43, treatment D had a mean of 3.50 and treatment E had a mean of 3.57 all interpreted as with slight bangus-tatse

It only shows that regardless of the amount of bangus and the amount clam added to the formulated product the taste is still similar.

Table 3. Results of sensory attributes of different treatments in terms of taste.

Treatment	Replicate	Total	Mean	Verbal Interpretation
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	1	2	3			
A	3.8	3.7	3.8	11.3	3.77	With slight bangus-clam taste
B	3.4	3.6	3.7	10.7	3.57	With slight bangus-clam taste
C	3.2	3.6	3.5	10.3	3.43	With slight bangus-clam taste
D	3.1	3.7	3.7	10.5	3.50	With slight bangus-clam taste
E	3.7	3.6	3.4	10.7	3.57	With slight bangus-clam taste

In terms of Texture

Table 4 shows the result of the sensory attributes of different Treatments in terms of texture.

As shown in the Table, Treatment A and B had a mean of 3.53, Treatment C had a mean of 3.57, Treatment D had a mean of 3.23 and Treatment E had a mean of 3.57 interpreted as Slightly crispy.

The finely chopped parts of the bangus and clam, combined with the sizzling oil, create a crispy and crunchy exterior. It only shows that regardless of the amount of bangus and the amount clam added to the formulated product the texture is still similar.

Table 4. Result of sensory attributes of different treatments in terms of texture.

Treatment	Replicate			Total	Mean	Verbal Interpretation
	1	2	3			
A	3.6	3.3	3.7	10.6	3.53	Slightly crispy
B	3.6	3.4	3.6	10.6	3.53	Slightly crispy
C	3.7	3.6	3.4	10.7	3.57	Slightly crispy
D	3.4	3.1	3.2	9.7	3.23	Slightly crispy
E	3.2	3.5	3.4	10.1	3.37	Slightly crispy

ANOVA result of acceptability of different treatments

Table 5 shows the results of significant difference in the acceptability among treatments

Results shows that the significant value was .529 higher with the significant level of 0.05. Results shows that there was no significant difference in the overall acceptability of bangus-clam sisig.

Regardless the amount of bangus and clam sisig. All treatments are acceptable.

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Table 5. ANOVA result acceptability of the different treatments.

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	8.200	4	2.050	.796	.529
Within Groups	373.300	145	2.574		
Total	381.500	149			

Most preferred formulation of Bangus-clam Sisig

Table 6 shows the result of the most preferred formulation among treatments. Treatment A had a total mean of 2.83 ranked as 4th, treatment B had a total mean of 3.13 rank as 5th, treatment C had a total mean of 2.63 ranked as 1st, treatment D has a total mean of 2.70 ranked as 2nd and treatment E had a total mean of 2.77 ranked as 3rd.

Results shows that the most preferred formulation is treatment C (50% bangus. 50% clam). Bangus sisig typically uses flaked fish, onions, and chicharon (pork rinds) for a mix of textures – soft fish, crisp vegetables, and crunchy pork. Clam sisig might have a chewier texture from the clams. Having both dishes together offers a more interesting textural experience in each bite (Manalo, L. 2022, April 21).

Table 6. Most preferred formulation

Treatment		Replication		Total	Mean	Rank
A	3	3.1	2.4	8.5	2.83	4 th
B	2.8	3.5	3.1	7.9	3.13	5 th
C	2.7	2.8	2.4	7.9	2.63	1 st
D	2.5	2.6	3	8.1	2.70	2 nd
E	2.7	2.4	3.2	8.3	2.77	3 rd

Consumer Acceptability of the most preferred formulation

Figure below shows the results of consumer test consist of 50 college student and 50 employees. College students had a mean of 8.22 interpreted as like extremely, employee had a mean of 7.8 interpreted as like moderately.

Results show that college student more like the bangus clam sisig that employee. College student consume sisig more often due to their lifestyle. They may frequent street food stalls or eateries (Sali et al., 2023). Sisig is often cheaper option compared to other restaurant dishes, which might suit college student on a tight budget (Sherman, 2022)

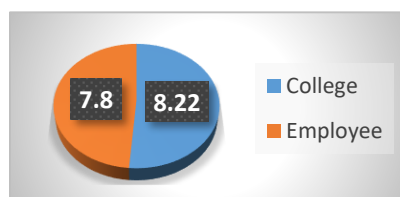


Figure 1 Consumer Acceptability

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(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

The study aimed to evaluate the acceptability and sensory characteristics of a newly developed bangus-clam sisig product. Milkfish (bangus) is rich in omega-3 fatty acids and protein, essential for the body's proper functioning. Clams are abundant in Vitamin B12, iron, protein, riboflavin, choline, and other essential minerals and vitamins while being low in fat.

Five different formulations of bangus-clam sisig were created, each with varying proportions of bangus flakes and clam meat. Sensory evaluation was conducted to assess color, odor, taste, and texture. The findings indicated that these sensory characteristics remained consistent across all treatments, regardless of the proportions of bangus and clams. This suggests that the varying amounts of these ingredients did not significantly impact the dish's sensory qualities.

A notable observation was that during pan-frying, the skin browned and crisped up, contributing a significant amount of brown color to the overall dish. Despite this, the overall acceptability of the bangus-clam sisig showed no significant differences among the treatments. This indicates that the dish is versatile and maintains its sensory appeal with different proportions of bangus and clams.

Out of the five treatments, the most preferred formulation consisted of an equal proportion of 50% bangus and 50% clam. This formulation was favored by most college students, highlighting its optimal balance for achieving the best sensory experience.

In conclusion, the study successfully demonstrated that bangus-clam sisig is a highly acceptable dish with desirable sensory characteristics, irrespective of the proportion of bangus and clam used. The most preferred formulation, consisting of equal parts bangus and clam, underscores the potential of this combination in creating a nutritious and palatable dish that appeals to a broad range of consumers, particularly college students.

The addition of bangus and clam in the developed sisig even of varying percentage had no intense effect on its overall results in terms of color, odor, taste and texture, however, during the cooking process, especially when pan-frying, the skin browns and crisps up. This contributes a significant amount of brown color to the overall dish.

The result shows that there was no significant difference in the overall acceptability of bangus-clam sisig. Regardless the amount of bangus and clam sisig. All treatments are acceptable.

Out of five treatments it shows that the most preferred formulation consists of 50% bangus and 50% clam.

It shows that most of the college student preferred the bangus-clam sisig.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

The study aimed to conduct product development on bangus-clam sisig. It was conducted at Capiz State University Fisheries and Food Research Development Center, Dayao Satellite College last May, 2024. The study used 5 treatments (5) with three (3) replications. Ten (10) panel of experts evaluated the sensory qualities of the products, and One hundred (100) panelist for consumer test (50 college student and 50 employee).

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(15) Technology Transfer Activity (Indicate the specific products, activity design, processes or services transferred as extension activities)

Production of Bangus-Clam Sisig. Process will be taught to the target clientele in the community through extension program of the college. Present the technology to the community and entrepreneur. Present the study in research symposium. Bangus-clam Sisig for family consumption and backyard micro industry.

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.)

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

First and foremost, the researcher would like to extend his sincerest thanks and praises to Almighty God for his provision of life, good health, guidance, strength, and wisdom which sustain the researcher throughout the study.

The researcher is also indebted to the following persons whose contribution, assistance, and guidance helped to make this research possible:

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